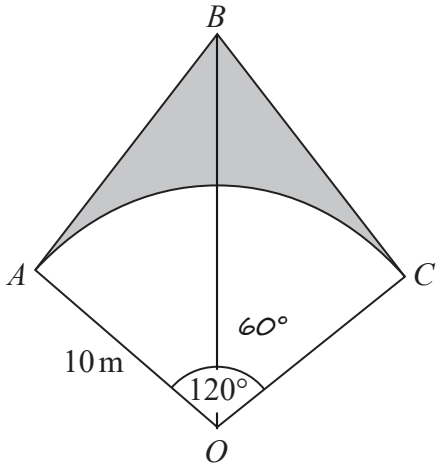
20



OAC is a sector of a circle, centre O, radius 10 m.

BA is the tangent to the circle at point A.
BC is the tangent to the circle at point C.

Angle $AOC = 120°$

Calculate the area of the shaded region.
Give your answer correct to 3 significant figures.

OA is perpendicular to tangent AB, so angle OAB = 90°.

1 mark
Radius meets tangent at a right angle

The two right triangles are congruent, so angle AOB = 60°.

tan 60° = AB ÷ 10, so AB = 10 tan 60° = 10√3 m.

1 mark
Find a tangent length using trigonometry

Area of kite OABC = 2 × ½ × 10 × 10√3 = 100√3 m^2.

1 mark
Add the areas of the two right triangles

Sector area = (120 ÷ 360) × π × 10^2 = 100π/3 m^2.

1 mark
Use the sector fraction of the whole circle

Shaded area = 100√3 − 100π/3 = 68.4853... m^2.

1 mark
Subtract and round to 3 significant figures

68.5 ..... m^2

(Total for Question 20 is 5 marks)